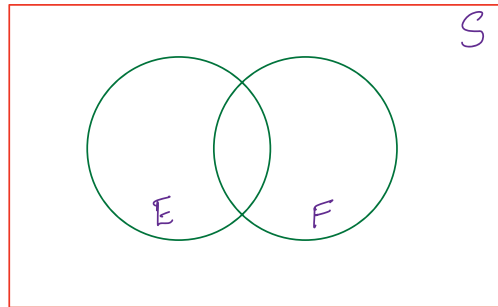
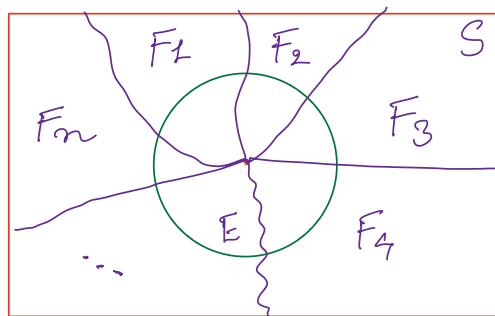


Law of Total Probability



$$\begin{aligned} P(E) &= P(E \cap F) + P(E \cap F^c) \\ &= P(F) \cdot P(E|F) + P(F^c) \cdot P(E|F^c) \end{aligned}$$

Further Generalization :



Suppose sample space S is partitioned into events $F_1, F_2, \dots, F_n$, s.t.

and

$$F_i \cap F_j = \emptyset, \text{ for } i \neq j$$
$$\bigcup_{i=1}^n F_i = S.$$

Then, we can write,

$$E = (E \cap F_1) \cup (E \cap F_2) \cup \dots \cup (E \cap F_n)$$

$$\begin{aligned} \text{So, } P(E) &= P(E \cap F_1) + P(E \cap F_2) + \dots + P(E \cap F_n) \\ &= P(F_1) \cdot P(E|F_1) + \dots + P(F_n) \cdot P(E|F_n) \end{aligned}$$

Bayes Theorem:

for a sequence of events $F_1, F_2, \dots, F_n$ s.t.

$$F_i \cap F_j = \emptyset, \text{ for } i \neq j$$

$$\text{and } \bigcup_{i=1}^n F_i = S$$

$$\begin{aligned} \text{Then, } P(F_j|E) &= \frac{P(F_j \cap E)}{P(E)} \\ &= \frac{P(F_j) \cdot P(E|F_j)}{\sum_{i=1}^n P(F_i) \cdot P(E|F_i)} \end{aligned}$$